

A Packing of 80 Lines in $\mathbb{R}^{15}$

Limin Ge*

October 8, 2026

Abstract

We give 80 1-dimensional subspaces of $\mathbb{R}^{15}$ whose smallest squared chordal distance is at least 0.915616. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.914982, obtained by deleting 1 of the 81 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 80 subspaces of dimension 1 in $\mathbb{R}^{15}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.915616 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 15$, $n = 1$. The cell $N = 80$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.914982, obtained by deleting 1 of the 81 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(15 - 1)/15 \cdot 80/79 = 0.945148$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m15n1N80.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 80 seconds on one laptop, with about ten such searches running at once.

A Exact value

```
numerator:
915616579032221802106005813356450233815527054324
denominator:
999999999999139340615138184153729850106294921085
```

B Coordinates

80 blocks of 15 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m15n1N80.txt`.

-131613895173	344924706573	-551575185312	-459946627383
-144918197227	-122354247133	-66980865026	206690866087
-342452419052	-91721248919	444724429625	169107456561
-335039215961	-16698950984	74398693750	-176026755038
158790465503	-95895961750	-231252430362	200375201940
231382901834	-249123631190	-78402876363	-26779225222
-412320454870	-232016682439	-291889098421	32616422619
88993751014	226799932687	-76678746845	259192785128
181638926963	-39114430675	31075687019	333330191438
-214858700837	366400090918	56731283115	-76365980937
-548276331924	198099377259	-121093882133	37361568887
24306470814	133209741565	-134404095649	-26546638836
69826981202	-120969277809	538364507814	164043544520
271465407027	-210991224423	-282416467813	-428726190989
128635463396	52299617773	-190395065936	-330276457253
138105534890	-548203089388	-144569393899	-274159277655
-55862101336	58396134813	-346098716498	314706728599
-225570424840	-140401416847	83969687097	-162515126350
-167015235378	-193247595691	-27709896439	284836890173
183632461467	465854335935	29148978066	129068473274
-505221013602	224046781749	438681052751	-54709423641
80526344718	-64501122030	-281908485774	-92337622614
-10718320861	-415219455930	-287996512635	86867967912
12371865631	-155906246556	93248538794	332123781101
-125055375684	183421070563	-220277875797	-533493507281
-176016924046	323708481692	311335202347	37651340132
273356693030	92076488592	124585962029	-105095880838
124996929272	514339316260	277240148697	271464015981
-658420714915	143249588651	-377839009393	238061780243
-181167971763	-422579322896	-445649052686	380497902589
-264178202486	173163919084	-90123618478	469890775166
24003575656	3278544371	158994663112	-265261277008
193290552610	300799308135	408358465709	-418675332193
162912009037	-108714788706	136283363672	-290203933516
26525239855	-78011914011	-144828795656	217367035124
-13647515266	-178744150136	302050877495	-399647743663
476600396861	250298390472	48565655331	-324736232922
-360644790277	-203617902979	92035407707	-93790200094
-22264113470	-11573890975	-260546845047	-90067123618
200966533368	288747696624	244266172693	186273404389
-98186109036	444286486583	-142613898158	209355717539
-117789600108	-453578375098	135941004097	-168304823042
543484406656	-218448979218	-390227588851	-1752271950
384292991549	-116941951441	465498789904	111160456526
-13234628432	-272527445791	-209237762	28086874567
-377808815171	-4982008983	228145602501	567740869391
62621042390	-73284427439	140106588981	14763001461
-208403971520	106277359360	216329506946	123777298470
-9197156319	-472934797669	5388758440	380553829571
399779685826	-35829728112	-38421386257	-108478849934
-289902377792	144019992284	223569025155	49707419745
597990158717	235431117916	331130970457	143952121062
192624402030	310827916071	-248865526621	564350312331
-92176679144	-135148090525	-249808998215	-354275068608
364985098891	295454520848	-433890766448	-28277579304
4403336284	301273329788	114396812388	-128972620448
128268182129	86711774687	323810093797	-52431070283
-68487259637	205707789035	72608002222	-73118120465
-60082489027	-250440690897	389995901850	-82419031425
-69105256260	107479176831	95154402579	-77625959445
-94286051515	-57090237941	60993262775	-439440728933
-171391270011	522016486198	291070677519	-93015341286
-172111925168	295337470331	432920067235	-207891025717
456994075173	-379178335532	412382056509	487062936783
-146908881032	-4955757215	283876302412	-33392316175
-86095522349	-170331196422	161146179549	-12007233749
-556720424945	216006149532	141632570330	182943537366
15930364214	-244323675222	366973110107	-364849474092
-476361511543	-444095325622	91831216241	59074180779
-73016604648	22126787723	614071115	35656254365

136636878093	68519152812	200083919022	-87959430128
-58186234596	406502855973	259066095667	242474647871
-76738860503	114738126179	-266048099272	-276551630895
-441327576785	-82573421327	-145153528035	71263996315
349961465913	-376580649972	460656795575	-42012151053
-208527777820	-13884216764	-262340384761	-7351636696
92642949243	49554543567	586147354226	46282150945
116703061008	-78213928757	-55159064173	127250594136
-39623444219	-108571202140	-263601258887	166303563404
-468743250233	131181428134	-17846929101	-567914165126
28938458000	-562107674974	-6698886935	-444805341302
-197213347712	-32377090175	-59364125263	-147227458369
-143362325656	-311828513375	-253579804255	29617137249
121628458166	16924379607	457082388128	-451294302964
12020638595	61352995372	-133858547744	24224344947
-220363429582	347186191580	283413249701	-361652943545
488669303333	-132759698589	-232813678318	-585307421527
213865126783	217906106346	381397780533	-181585247220
263447802934	106085370682	423435848249	-108713055441
485078915331	-83524670506	-128298077022	-61919587935
-334648606012	342316155836	-286083580619	315584090680
-279738152212	389647017581	-148638815662	135034073633
-401625189255	282525539796	-164936916375	-185781812810
-102136039177	-261308745143	295339174854	-99124091197
-73990033006	-391256421866	-102814895977	-299324422001
-221518906384	-350424115751	-25402225444	-7250711728
-281828351942	38771451380	340689654973	-158713153492
-25758062015	186778207780	262886208601	156111797571
-98626176565	-140150079707	-332042018726	-419563294896
561737582159	-73530197691	441795584604	-11249407284
-279728732656	-58194393690	131053383034	-88044726501
160243504179	18946552270	97213984069	-82417163330
-5441273255	-121347253481	17575583306	239246640108
-212392538583	-105575549208	-292913652895	245683521300
170422676432	-478726180838	-113547946751	-30261874620
54275481734	555752675674	38935605614	35689042086
-8481430262	-163502174542	207348408388	-175520504975
-34776284180	-3371404289	-490909834149	543793868602
210826704571	-519314983152	257042799766	363302942865
-256334167179	414255407517	165171439002	-129417764158
161599978080	80776294026	70041073429	42743800929
-276801377296	112697104837	-212906645300	-176402392237
-91443229570	117500245961	274208776168	310579796484
-211109929982	-23836675682	122644372543	510762469529
-85342324782	73553445145	27084103536	37168807481
-625417734955	-226722471959	-474372002966	-313964239664
409506609292	-263560901725	-345036647092	-171915823205
-155066832894	-236101252266	-158328516385	9514408818
-297834444201	234896290077	-280039295520	87917079073
-226003369958	-467766182052	140998577321	-245943713252
207056392719	-212431278931	-460065492826	170279503920
-106134960481	82075429388	139358760286	-434302748946
-260502505010	-131294448581	175615774829	63015998343
-87263312311	188692911583	-153391210007	70604189166
340679691159	-70187613317	-328964444932	20085100068
226872981051	-334851976432	-50862709703	672896539469
110053172545	-164243177168	-312414365848	-318589813243
-208900466174	-67432691794	-155297806458	116081383899
321844411754	18442153248	291185472489	-86279564224
376375154154	375564432244	-170727532012	310759942825
320056046842	270690070262	-218503751905	37850619087
-17799563623	-2396293426	230720071673	139252219855
295085707631	195150202295	317518326343	181564828014
-456920280545	-222610245055	47113585949	688866964018
57052390872	-342961608442	78753714830	-332907362764
90691953743	-236539665792	594761818726	140514879055
401644829543	-478119015811	202367933272	-226744892642
235707836084	333679250665	52964024976	105980832717
-22679542766	-301645729023	25047393980	-127576332910
346047555213	-191765606591	-112646724826	99341864896
165234945098	166715983608	378564886908	207277762551
45929623008	-346317055240	-355178430137	300299702668
-348816915350	343867579241	177845298516	111070684499
294654796295	575286094056	226546305942	101285705910
401128206745	240271802183	209193000525	-164420833018
161946065569	-152920116869	-69640035127	348577461353
-163707659252	-200178692179	-308532011492	-178099573682
-119257019457	-187121853498	-99105395155	-467336082687
227541479936	61296102184	357420098845	-164735524558
-369865126314	31922211119	-330401517822	236084397032
-460415641737	39165186097	100421088884	-241627253852
134269590169	114330413756	-168740855296	36808658935
39043391049	-315791528288	-110683820092	-126735466387
389469254006	90414969818	-236405121115	-423598136666
-254854259313	90549983238	-540916318677	341646538546
-23956277997	-254801616924	32708743373	167968742493
-481505895304	131642585330	-434019838308	-133840698681
-64971566973	324509813410	-460608700295	307214612065
369947780136	-304418320995	-94593728358	77443927817
183206522803	-214799614648	132072273693	-77303096473
-158650305598	414816944734	266047984156	119329047868
-197873704741	-218383236504	161362777364	-217420897269
202861279558	101311466750	97361480977	-549001942399
-76101859779	-196549445969	160765215121	-12377524111
-183004084658	492945946981	206309348886	-7403837648
-401155240250	-104877425138	-38209265522	143767170445
149408633651	341734450001	117448283212	-104630647364
-407475704233	133486864533	470062142390	-81680535529
27988155654	-66449172366	342445626818	411790296090
114943049222	-206568481336	81490573790	-91663648742
471306082336	-23167082339	195564375054	-154273590613
272192829865	236410419059	-239794058434	-454089136382

402580581908	-46769602551	701624243313	529302838106
150238633161	82733438025	181639290134	150287906928
142455356515	-339292507458	258184012293	14774377123
-211954304335	546400949082	221049935919	271910915894
-284812499627	-269283654535	-218823933180	-263513706526
195827127691	122002067319	-14572543555	16085978935
70385409561	430450986744	-107654008489	231983852099
-117883106080	-501465196361	-83489307336	100670701684
78651022197	-115767965657	-339631830395	-229442425822
-5971411868	201558425588	-353917565559	112090018033
-388485427812	-164144668148	189322689047	-26750916918
278366351988	319718627013	110870287012	135781702503
178790020539	-247592129362	331780778381	-491575018584
314135331509	275031857074	-475784304221	79224676701
546419093524	-260240996219	-497482088857	34132166578
2503679159	273178660060	27652405445	-317714386191
-364153156097	-2649683302	138327077271	291731182240
4160288579	-272431699134	-172164556100	15291785285
406555124696	-103794459965	-87986683367	268240769105
136611070850	29545292249	-17254248133	-133774165822
200654284399	369807720921	-138295681821	-238859527708
54772104658	-13800577866	28592411846	351880815651
-158056961084	93083027201	223153311195	347064321767
347198095354	-23694912870	-326356035974	-144278763716
173172490657	142428381653	-205381834120	287268588482
-284883280488	-201063905825	145530454157	-227324782002
62688152188	-133402953591	-290135013394	-33633598561
-6974644242	30501110118	-381666336778	-336783234844
-135506700746	715751340526	42915947888	30847467354
-187473037945	-131170197064	42634203032	-20182280232
199679547992	-61226004917	252543897398	-232113098123
647098616398	-443104626312	413907825492	-15683931520
36840042984	167997455147	-215202369065	314293364248
-475409314845	-62053365597	16628009145	31972827713
143048005039	-112769633149	-154498494015	-602977977052
147421222500	-276046645285	375042382956	104425292219
-282394646076	-268217807756	362367526161	131903961100
-275279167772	450359636258	-138204552358	-561937346406
-179750083994	-25744815260	191382794939	32076004842
-303638900972	-35791763505	-107442215874	130403235249
268305611853	-48265966216	347400504126	66848290875
210374182342	232968201977	-94317413275	-98425812218
401751943378	-317540807607	10916983303	4664939527
12319408558	-123415222141	625724146487	-61900016886
390685651442	-373485182137	-28234842185	-26613388592
44590989162	324046358752	369588723401	-274315300727
134073824595	134221747544	394501393974	-195017172571
518165937740	-132993115788	170910992264	186569727043
-175568898223	178126510685	3064517485	-64626002111
25610541365	403104970406	-62616234427	611744771521
-235751398384	259558087170	-174742307928	372404938106
159384673185	129916976108	-173094304389	13645925729
288797461968	-266033053794	-209605293203	16756678411
79538117497	217267113569	-555760813992	-427652897638
-174350557688	-432945840512	-445301297035	-279699540491
-147638921574	220744122435	174699299201	-242862952092
-172450897990	89724901542	289855174537	-43693097424
-461200606314	264536649061	-143125668323	199501964017
-239101265451	451436404964	-267565358155	-28877605573
276273335280	-240934311934	191833467730	-69456823440
-283013128024	-283218170327	-44347716524	-142697690250
-145181816636	95879769503	-100759571137	-234117355226
272552710955	-199997252530	-71111943922	411436210224
-312626861046	302439473220	60869055485	188958162806
129275668025	-29226028060	6078838506	-9112693453
237713540542	127557581201	-32092250235	-415735996447
-113915879466	386912404268	-116273839962	7330007713
313969472374	-11192860190	-476803224505	595908119054
329967683558	-338220427302	355501702599	284787142491
81921960996	-36844845439	-257936455616	183483440748
165706084368	-403376847600	123312442254	-172115191429
-153005824940	78050043863	344500508099	33725420045
-3940877149	-159858684815	26460337927	-791228126858
-232815272166	71044350906	-123669431847	58000966304
337234453100	472846992013	-414336137194	-306868856306
33999318634	279739008945	40579413766	194848243827
554657734451	384819679316	233404296754	-2458775698
983124841	234647015733	238341095868	224908665423
310734149258	52728566681	-65187626107	-87662906555
-106783897877	104791626397	72767400910	309149029916
-132462721380	278855593504	-396730215447	26135573570
124235610123	165750482370	287334851091	15995143683
264106636174	503547961223	-350856978126	-184786618033
-193769537296	47199633428	-216470482482	-210899604213
331021139940	-450017360523	-297640622477	-22715886070
-277509054420	-100849514840	-355975863338	58263876581
478578096908	52659878335	-353490174459	102120663399
555427565627	-156443073933	-257909494238	-137592465083
-56815492609	253352989131	-181863056487	102823532162
-118997699406	148721108698	158907621373	303915284242
-86976031364	-204644273528	-38214664471	132875098995
-51811044225	168911712281	-118711457955	-202916710159
-207251418142	306439277006	41575812811	374402537614
-33977384747	-49076119569	596770052943	299771941590
-17114320620	-382118550746	98525958580	342026969980
-69463264243	-95694119679	-117839292528	-304126517654
408461855025	-436426435217	-298076536984	38582318891
-116366719327	-140395021385	-46631108438	299982641322
168144578261	159168009462	-191127523860	-360110613712
146646620230	-81778055229	-64671341523	-254709522458
386170302397	-168358111413	413448726536	288653567910
-276165533395	-113356469605	460296990445	254631092334

27406453435	220313286269	4917810627	-277966432254
77186358381	301719528337	8502968997	-289063618090
12694696786	-311005997505	182473521261	452004968037
180624370366	264316868426	191357092612	108993386857
-267757209041	63243175224	505932104489	78107487993
-61651957968	367050919960	-114314524582	-218778020197
-95126502917	-229237498076	81944686415	-189860533258
-150231231766	451162231864	549027231479	-360122107384
-506049023458	-315944862550	-361435775972	171656245644
183439691821	-38244211555	-302152805291	-224341702868
-407575078957	326821573620	299624740830	-293246668373
-25198687950	134343928204	-292828913971	110413046140
-409809567914	399011580359	-36095390072	-62725336789
405120283060	-13275712344	117753842677	133779694861
11306688356	-71661118062	-252190807713	-440560582254
182818289594	-321811473433	103182630724	43164627584
223878449878	-117206685575	-166587407008	-455962637387
239676723491	165369540394	148553218021	308172643393
178911164668	-49025575838	-58471520266	-406938426882
-553816661316	44904412916	-179222939293	3860233728
-146595939696	-61186880311	-329504753732	-178466920141
186467356055	-229657401252	-133285083384	128719449800
-232025579122	-36439432577	-54814126565	-119820579464
225192962617	-134648267635	290198070467	65594530457
10910128948	-683059214171	-209025785862	150049882116
-350622984155	327574735508	326480197033	-460370355518